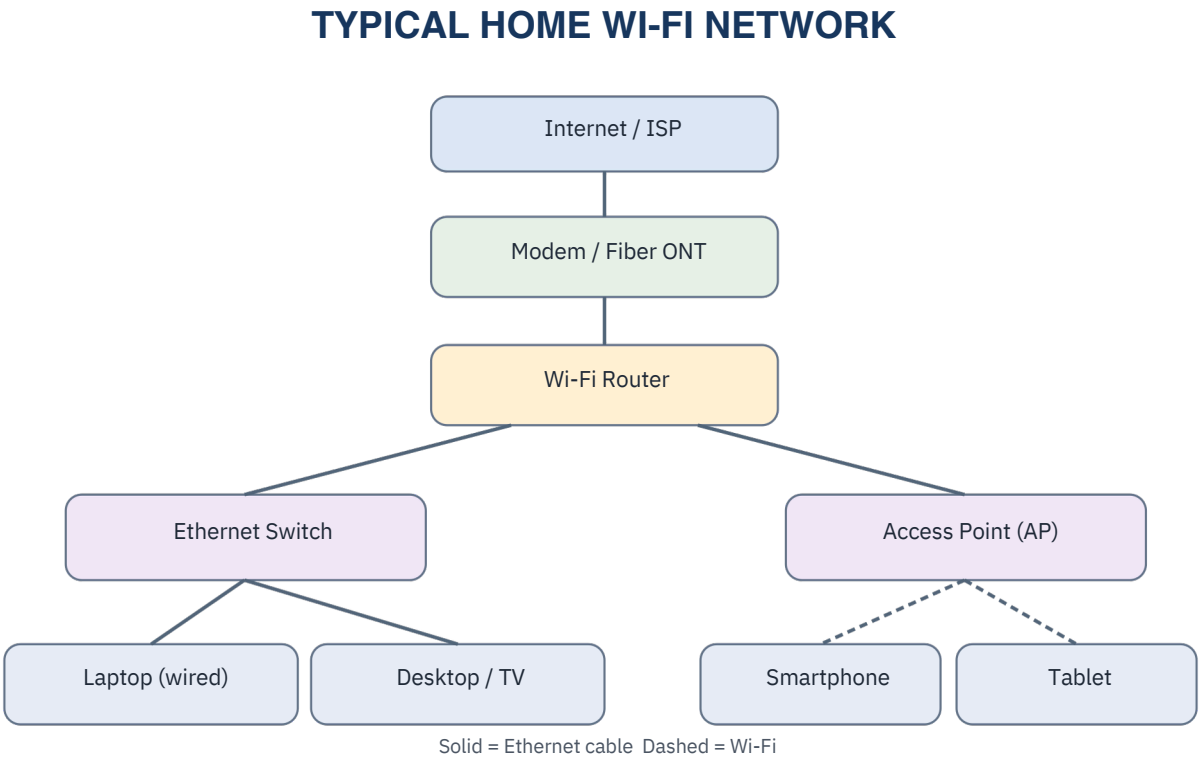
Task 1: Labeled Diagram of a Typical Home Wi-Fi Setup

A home network usually startswith the connection suppliedby an Internet ServiceProvider (ISP). The modem, or a fiber ONT in many fiber connections, passes the connection to the router. The router manages traffic between the home network and the Internet. A switch adds more wired ports, while an access point provides Wi-Fi coverage in another area.



Task 2: Main Functions and Real-Life Examples

	Main function (in simple words)	Real-life example
Modem / Fiber ONT	Connects the home to the ISP's access line and provides a network hand-off to the router. A fiber connection commonly uses an ONT rather than a traditional cable/DSL modem.	The ISP's device near the fiber entry point or broadband line.
Router	Connects different networks and sends IP traffic toward the correct destination. Home routers commonly provide Wi-Fi, DHCP, NAT and basic firewall features.	A home router lets phones, laptops and TVs share one Internet connection.
Switch	Connects multiple wired devices on the same local network and forwards Ethernet frames toward the correct port.	A small office uses a switch to connect desktop PCs, printers and a network storage device.
Access Point (AP)	Adds Wi-Fi access to an existing wired network. It is useful for extending wireless coverage; it normally relies on a router for Internet routing.	An AP mounted in an upstairs room provides Wi-Fi where the main router's signal is weak.

	Main function (in simple words)	Real-life example
Network cable	Carries data between network devices. Ethernet twisted-pair cable is common for LAN connections; coaxial and fiber are used in other access and backbone systems.	An Ethernet cable connects a desktop PC to the router or switch.

A quick example from daily life

Ifaphone connects to Wi-Fibutthe Internet is not working, the Wi-Fi link may still be active while the ISP connection or router's Internet service has a problem. Checking the modem/ONT status lights and the router's WAN connection is a sensible first step. This is why knowing each device's job helps with basic troubleshooting.

Task 3: Three Common Types of Network Cables

Thecabledependsonthejob, distance, equipmentandserviceavailable.Thecabledrawings below are simplified illustrations, not photographs of a specific product.

1. Ethernet cable



Usually has an RJ45-style plug. Commonly connects computers, printers, switches and routers in homes, computer labs and offices.

2. Coaxial cable



Has a central conductor and shielding. Commonly used for cable-TV/broadband services and some CCTV installations.

3. Fiber-optic cable



Carries data as pulses of light through glass or plastic fiber. Commonly used for ISP fiber connections and high-speed links over longer distances.

Task 4: Router vs Switch — Comparison Table

Point of difference	Router	Switch
Main use	Connects different IP networks, such as a home LAN and the Internet.	Connects devices inside the same local Ethernet network (LAN).
Typical location	Usually at the boundary between the home network and the ISP connection.	Usually inside a LAN, near the wired devices it connects.
How it forwards data	Uses IP addresses and routing information to choose the next network path.	A typical Layer 2 switch uses MAC addresses to forward Ethernet frames to the appropriate port.
Ports / connections	Often has a WAN/Internet port and several LAN ports; wireless models also provide Wi-Fi.	Usually has many Ethernet ports for PCs, printers, access points and other wired devices.
Common extra features	Often provides NAT, DHCP, firewall features and Wi-Fi in a home model.	Managed switches may support VLANs, monitoring and traffic-priority features.
Simple example	Lets a phone on the home network reach a website on the Internet.	Connects a desktop PC and network printer to the same office LAN.

Conclusion

Ahomenetwork works because different devices handle different jobs. The modem or ONT provides the connection from the Internet service provider, the router connects the home network to other networks, the switch adds wired connections, and the access point provides Wi-Fi. Ethernet, coaxial and fiber-optic cables are chosen for different types of links. In a real home, several of these functions may be built into one device, but understanding the separate roles makes setup and troubleshooting easier.